Part A — Setup & Load

```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns

sns.set_theme(style="whitegrid")
```

```
df = sns.load_dataset("penguins")

print("Dataset Shape:", df.shape)

display(df.head())

df.info()
```

```
Dataset Shape: (344, 7)
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex
0	Adelie	Torgersen	39.1	18.7	181.0	3750.0	Male
1	Adelie	Torgersen	39.5	17.4	186.0	3800.0	Female
2	Adelie	Torgersen	40.3	18.0	195.0	3250.0	Female
3	Adelie	Torgersen	NaN	NaN	NaN	NaN	NaN
4	Adelie	Torgersen	36.7	19.3	193.0	3450.0	Female

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 344 entries, 0 to 343
Data columns (total 7 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   species            344 non-null    object
 1   island             344 non-null    object
 2   bill_length_mm     342 non-null    float64
 3   bill_depth_mm      342 non-null    float64
 4   flipper_length_mm  342 non-null    float64
 5   body_mass_g        342 non-null    float64
 6   sex                333 non-null    object
dtypes: float64(4), object(3)
memory usage: 18.9+ KB
```

Part B — Clean the Data

```
missing_values = df.isnull().sum()
print(f"Missing values per column: \n{missing_values}")
```

```
Missing values per column:
species               0
island                0
bill_length_mm        2
bill_depth_mm         2
```

```
flipper_length_mm     2
body_mass_g           2
sex                  11
dtype: int64
```

```
print("Shape before cleaning:", df.shape)

df = df.dropna().reset_index(drop=True)

print("Shape after cleaning:", df.shape)
```

```
Shape before cleaning: (344, 7)
Shape after cleaning: (333, 7)
```

How many rows did you remove, and what % of the data was that? Was dropping them safe? Why? Your answer: _____

Answer: Rows Removed: 11 rows (the dataset went from 344 to 333 rows),

Percentage of Data: Approximately 3.20%,

Was dropping them safe? Why?: Yes, dropping these rows was completely safe and justified. The missing data accounts for only ~3% of the total dataset, which is a negligible amount. Removing such a small portion avoids introducing artificial bias through imputation while maintaining statistical integrity across all features.

Part C — Categorical EDA

```
print(df["species"].unique())
print(df["island"].unique())

palette = ["#2b5c8f", "#d95f02", "#7570b3"]

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

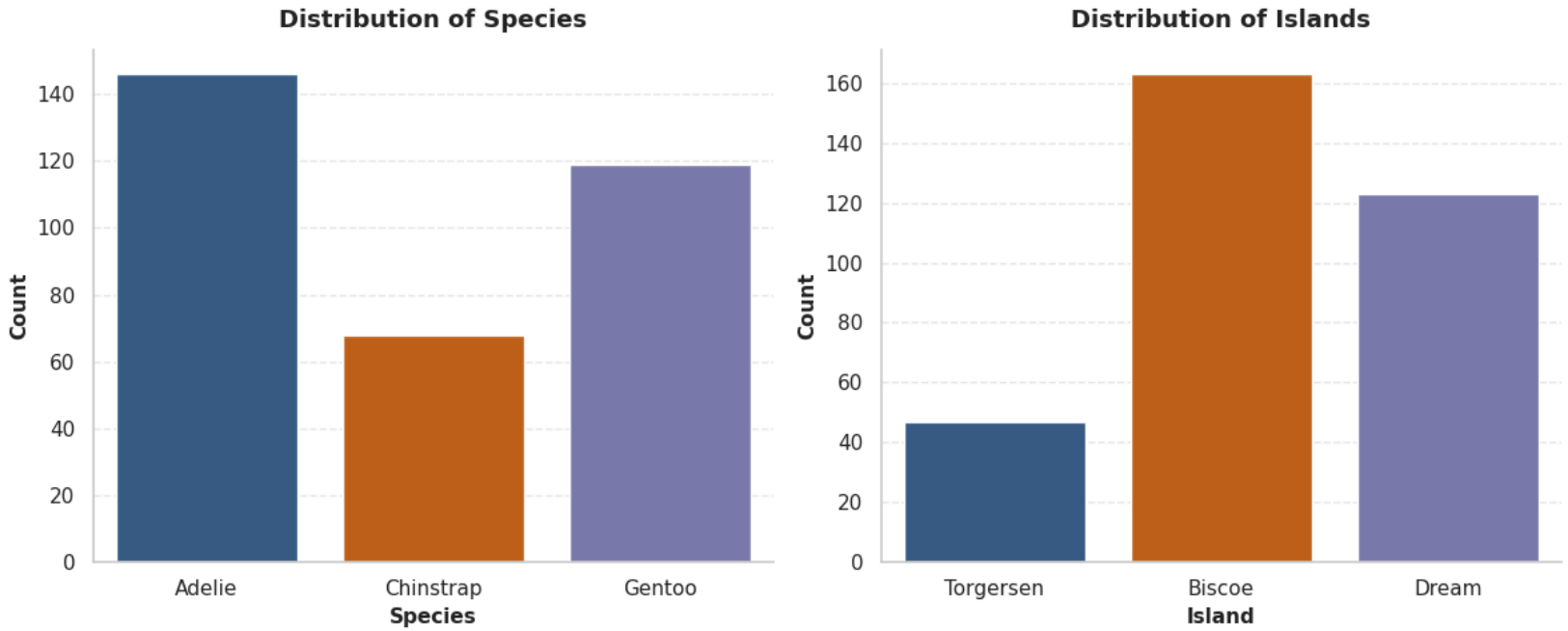
sns.countplot(
    data=df,
    x="species",
    hue="species",
    palette=palette,
    ax=axes[0],
    legend=False,
)
axes[0].set_title("Distribution of Species", fontsize=13, fontweight="bold", pad=12)
axes[0].set_xlabel("Species", fontsize=11, fontweight="bold")
axes[0].set_ylabel("Count", fontsize=11, fontweight="bold")
axes[0].grid(axis="y", linestyle="--", alpha=0.4)
sns.despine(ax=axes[0])

sns.countplot(
    data=df,
    x="island",
    hue="island",
    palette=palette,
    ax=axes[1],
    legend=False,
```

```
)
axes[1].set_title("Distribution of Islands", fontsize=13, fontweight="bold", pad=12)
axes[1].set_xlabel("Island", fontsize=11, fontweight="bold")
axes[1].set_ylabel("Count", fontsize=11, fontweight="bold")
axes[1].grid(axis="y", linestyle="--", alpha=0.4)
sns.despine(ax=axes[1])

plt.tight_layout()
plt.show()
```

```
['Adelie' 'Chinstrap' 'Gentoo']
['Torgersen' 'Biscoe' 'Dream']
```




```
crosstab = pd.crosstab(df["species"], df["island"])
print("Crosstab (Species vs Island):")
display(crosstab)

custom_colors = ["#2b5c8f", "#e66101", "#5e3c99"]

plt.figure(figsize=(9, 6))
crosstab.plot(kind="bar", color=custom_colors, width=0.75)

plt.title(
    "Species Distribution Across Islands",
    fontsize=14,
    fontweight="bold",
    pad=15,
)
plt.xlabel("Species", fontsize=11, fontweight="bold", labelpad=10)
plt.ylabel("Count", fontsize=11, fontweight="bold", labelpad=10)
```

```
plt.xticks(rotation=0, fontsize=11)
plt.yticks(fontsize=10)

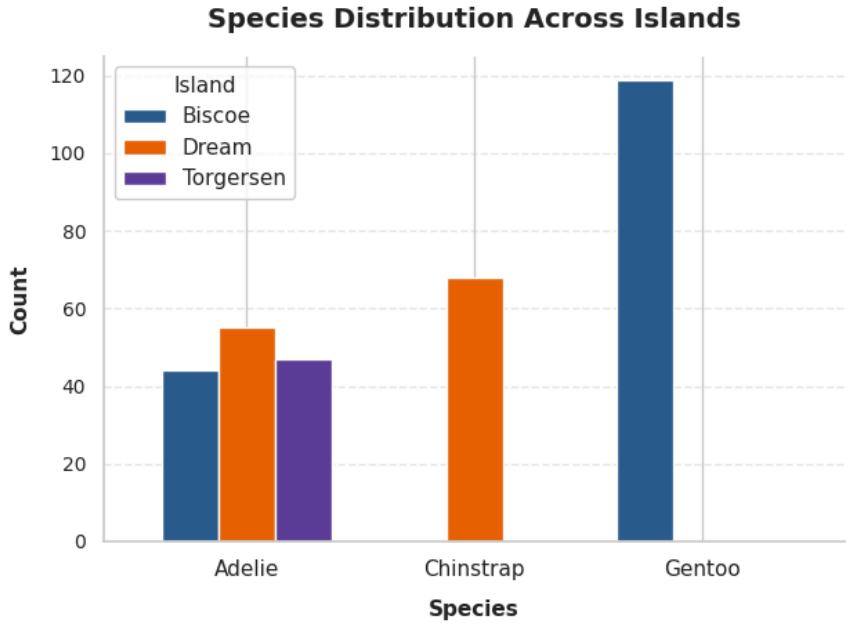
plt.legend(
    title="Island",
    title_fontsize="11",
    frameon=True,
    facecolor="white",
    framealpha=0.9,
)
plt.grid(axis="y", linestyle="--", alpha=0.4)
sns.despine(top=True, right=True)

plt.tight_layout()
plt.show()
```

```
Crosstab (Species vs Island):
```

species	Biscoe	Dream	Torgersen
Adelie	44	55	47
Chinstrap	0	68	0
Gentoo	119	0	0

```
<Figure size 900x600 with 0 Axes>
```



Which species is the most common? Is any species found on only ONE island? (That's a huge clue for identification.) Your answer: _____

Answer: Most Common Species: Adelie is the most common species in the dataset, accounting for the highest overall count (146 out of 333 cleaned observations).

Species Found on Only ONE Island: Yes, Gentoo and Chinstrap are both restricted to single, distinct islands:

Gentoo is found exclusively on Biscoe island.

Chinstrap is found exclusively on Dream island.

Part D — Numerical EDA

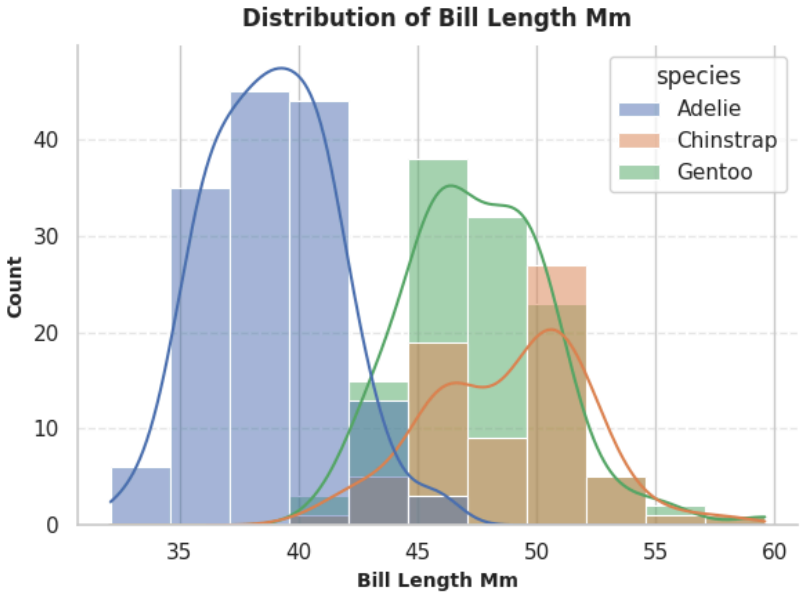
```
features = [
    "bill_length_mm",
    "bill_depth_mm",
    "flipper_length_mm",
    "body_mass_g",
]

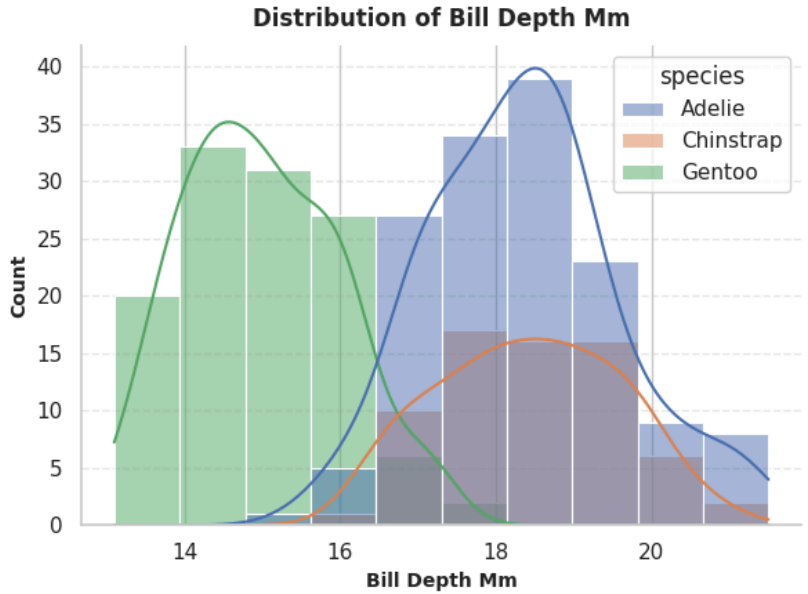
fig, axes = plt.subplots(2, 2, figsize=(12, 9))
axes = axes.flatten()

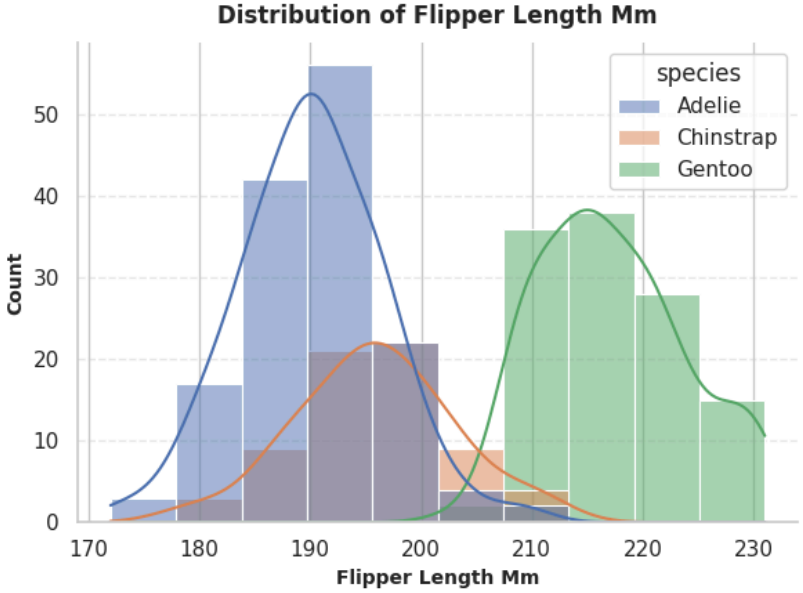
hist_color = "#256D7B"

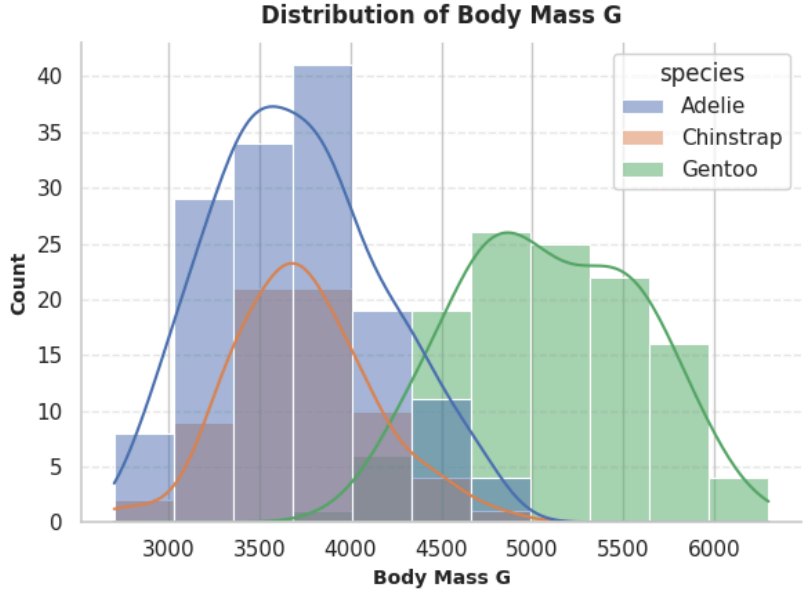
for i, feature in enumerate(features):
    sns.histplot(
        data=df,
        x=feature,
        kde=True,
        ax=axes[i],
        hue = "species",
        color=hist_color,
        edgecolor="white",
        linewidth=0.8,
    )
    axes[i].set_title(
        f"Distribution of {feature.replace('_', ' ').title()}",
        fontsize=12,
        fontweight="bold",
        pad=10,
    )
    axes[i].set_xlabel(
        feature.replace("_", " ").title(), fontsize=10, fontweight="bold"
    )
    axes[i].set_ylabel("Count", fontsize=10, fontweight="bold")
    axes[i].grid(axis="y", linestyle="--", alpha=0.4)
    sns.despine(ax=axes[i], top=True, right=True)

plt.tight_layout()
plt.show()
```







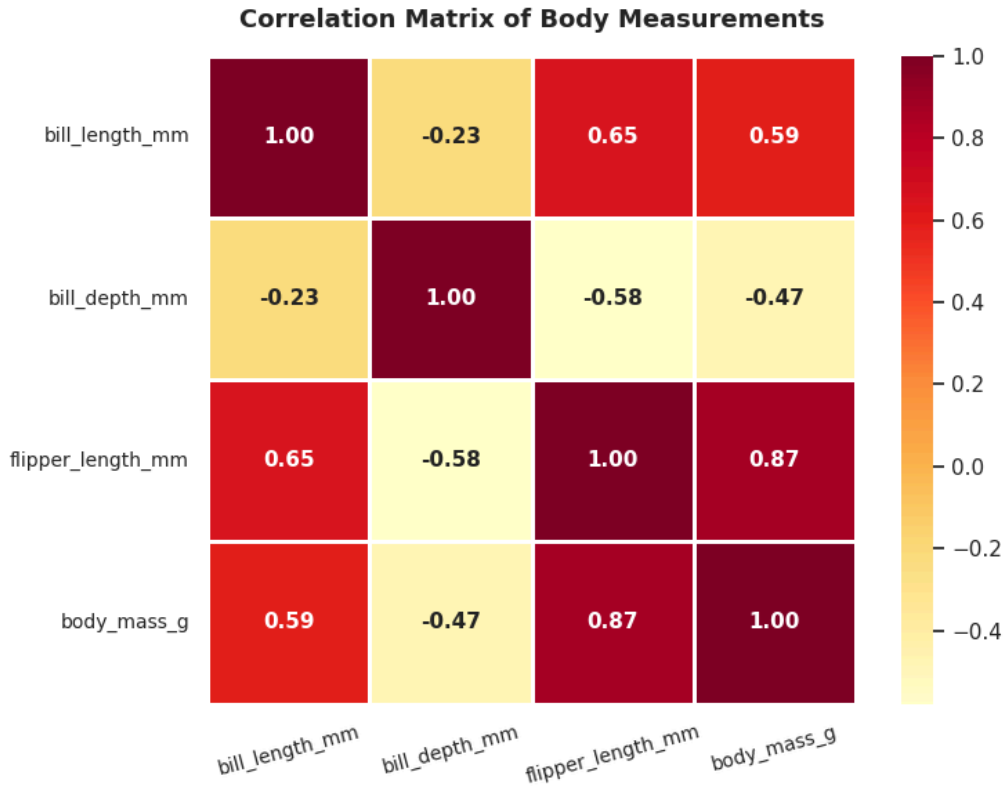



```
numeric_cols = [
    "bill_length_mm",
    "bill_depth_mm",
    "flipper_length_mm",
    "body_mass_g",
]
corr_matrix = df[numeric_cols].corr()
```

```
plt.figure(figsize=(8, 6))
sns.heatmap(
    corr_matrix,
    annot=True,
    fmt=".2f",
    cmap="YlOrRd",
    cbar=True,
    square=True,
    linewidths=0.8,
    linecolor="white",
    annot_kws={"size": 11, "weight": "bold"},
)

plt.title(
    "Correlation Matrix of Body Measurements",
    fontsize=13,
    fontweight="bold",
    pad=15,
)
plt.xticks(rotation=15, fontsize=10)
plt.yticks(rotation=0, fontsize=10)

plt.tight_layout()
plt.show()
```



Which two measurements have the strongest positive correlation, and does that make biological sense (think: bigger penguin → ?) Your answer: _____ Answer: Strongest Positive Correlation: Flipper Length (flipper_length_mm) and Body Mass (body_mass_g) have the strongest positive correlation ($r \approx 0.87$).

Biological Sense: Yes, it makes total biological sense. Larger, heavier penguins require longer, stronger flippers to propel and support their greater body size through water.

Part E — Species Fingerprints

```
num_cols = [
    "bill_length_mm",
    "bill_depth_mm",
    "flipper_length_mm",
    "body_mass_g",
]

species_means = df.groupby("species")[num_cols].mean()

species_means
```

species	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
Adelie	38.823973	18.347260	190.102740	3706.164384
Chinstrap	48.833824	18.420588	195.823529	3733.088235
Gentoo	47.568067	14.996639	217.235294	5092.436975

```
species_palette = ["#2b5c8f", "#d95f02", "#7570b3"]

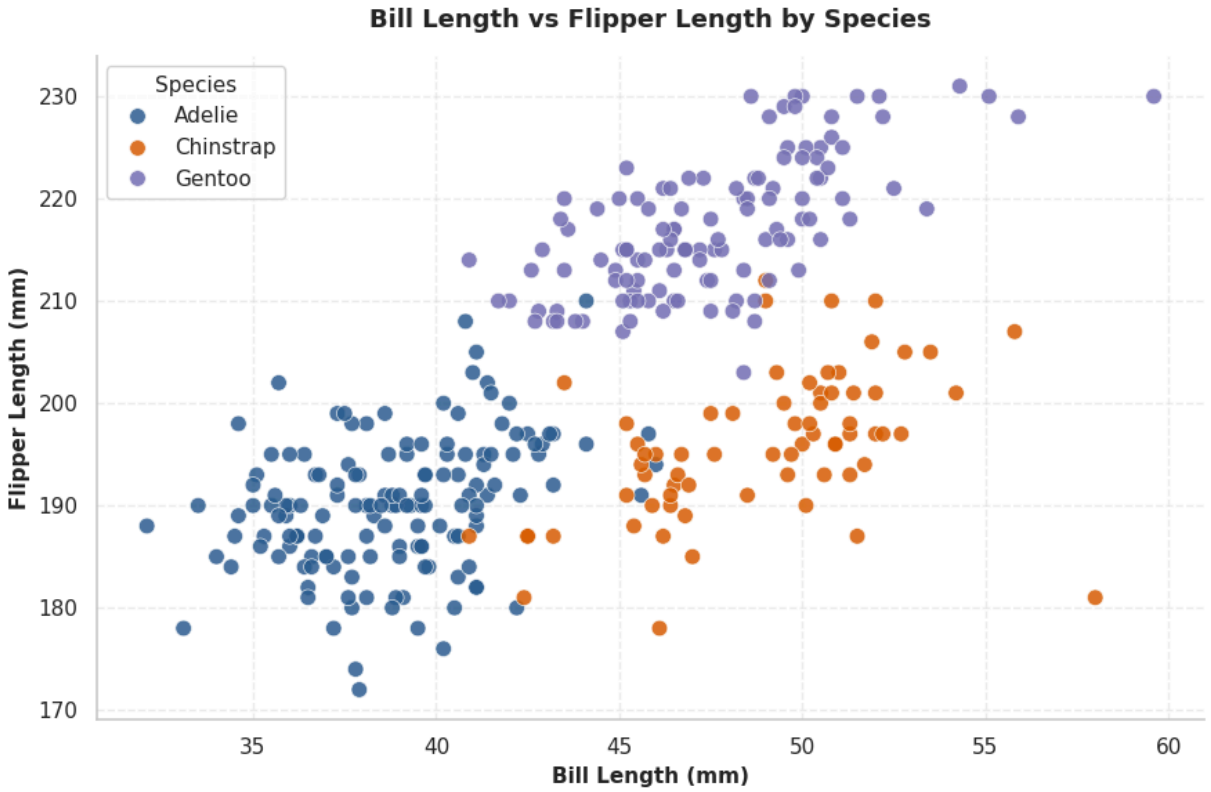
plt.figure(figsize=(9, 6))

sns.scatterplot(
    data=df,
    x="bill_length_mm",
    y="flipper_length_mm",
    hue="species",
    palette=species_palette,
    s=70,
    alpha=0.85,
    edgecolor="w",
    linewidth=0.5,
)

plt.title(
    "Bill Length vs Flipper Length by Species",
    fontsize=13,
    fontweight="bold",
    pad=15,
)
plt.xlabel("Bill Length (mm)", fontsize=11, fontweight="bold")
plt.ylabel("Flipper Length (mm)", fontsize=11, fontweight="bold")

plt.legend(title="Species", title_fontsize="11", frameon=True, facecolor="white")
plt.grid(True, linestyle="--", alpha=0.3)
sns.despine(top=True, right=True)

plt.tight_layout()
plt.show()
```




```
species_palette = ["#2b5c8f", "#d95f02", "#7570b3"]

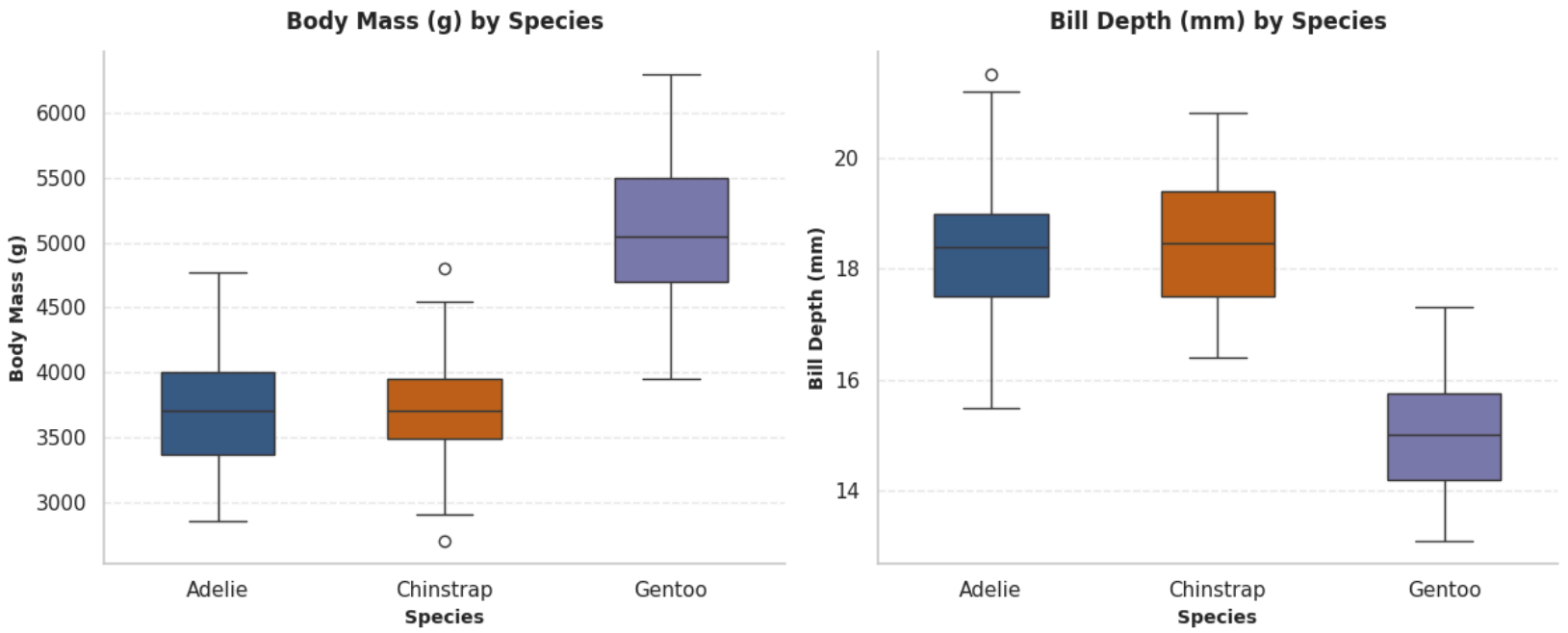
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

sns.boxplot(
    data=df,
    x="species",
    y="body_mass_g",
    hue="species",
    palette=species_palette,
    ax=axes[0],
    legend=False,
    width=0.5,
)
axes[0].set_title(
    "Body Mass (g) by Species", fontsize=12, fontweight="bold", pad=12
)
axes[0].set_xlabel("Species", fontsize=10, fontweight="bold")
axes[0].set_ylabel("Body Mass (g)", fontsize=10, fontweight="bold")
axes[0].grid(axis="y", linestyle="--", alpha=0.4)
sns.despine(ax=axes[0], top=True, right=True)

sns.boxplot(
```

```
    data=df,
    x="species",
    y="bill_depth_mm",
    hue="species",
    palette=species_palette,
    ax=axes[1],
    legend=False,
    width=0.5,
)
axes[1].set_title(
    "Bill Depth (mm) by Species", fontsize=12, fontweight="bold", pad=12
)
axes[1].set_xlabel("Species", fontsize=10, fontweight="bold")
axes[1].set_ylabel("Bill Depth (mm)", fontsize=10, fontweight="bold")
axes[1].grid(axis="y", linestyle="--", alpha=0.4)
sns.despine(ax=axes[1], top=True, right=True)

plt.tight_layout()
plt.show()
```



✍️ **Written Answer E: Describe each species in ONE sentence using measurements. For example: "Gentoos are the ___ (heaviest/lightest), with the ___ flippers and a ___ bill." Do this for all three species. Adélie: _____ Chinstrap: _____ Gentoo: _____**

Answer: Adelie: Adelies are the lightest penguins, with the shortest flippers and short, deep bills.

Chinstrap: Chinstraps are medium-weight penguins, with medium flippers and long, deep bills.

Gentoo: Gentoos are the heaviest penguins, with the longest flippers and long, shallow bills.

Part F — Your Field Guide

Write a short field guide (4–6 sentences) that a junior researcher could use to identify a penguin from measurements alone. Base every rule on your charts. A good guide gives simple decision rules, e.g. "If flipper length > 210mm, it's almost certainly a Gentoo."

Answer: Gentoo Identification: If the flipper length exceeds $206\ \text{mm}$ (or body mass $> 4,500\ \text{g}$), identify the penguin as a Gentoo, as they are distinctively the largest species with uniquely shallow bill depths ($< 17.5\ \text{mm}$).

Adélie vs. Chinstrap Split: For smaller penguins with flipper lengths below $206\ \text{mm}$, check the bill length to separate Adélies from Chinstraps.

Adélie Identification: If bill length is short (under $43\ \text{mm}$), classify it as an Adélie penguin.

Chinstrap Identification: If bill length is longer than $43\ \text{mm}$ paired with a deep bill ($> 16.5\ \text{mm}$), classify it as a Chinstrap.

Location Cross-Check: As a final validation, confirm location: a non-Adélie found on Biscoe island is a Gentoo, while one on Dream island is a Chinstrap.

🌟 **Bonus — Identify the Mystery Penguin**

A researcher measured a penguin but forgot to record its species:

bill_length = 46 mm, bill_depth = 15 mm, flipper_length = 217 mm, body_mass = 5000 g

Using your fingerprints from Part E, decide which species it is — and justify with the data (which measurements gave it away?).

```
# Define mystery penguin measurements
mystery_penguin = {
    "bill_length_mm": 46.0,
    "bill_depth_mm": 15.0,
    "flipper_length_mm": 217.0,
    "body_mass_g": 5000.0,
}

# 1. Compare mystery penguin against species means
num_cols = [
    "bill_length_mm",
    "bill_depth_mm",
    "flipper_length_mm",
    "body_mass_g",
```

```
]
species_means = df.groupby("species")[num_cols].mean()

print("--- Mystery Penguin Values vs Species Means ---")
print(f"Mystery Penguin: {mystery_penguin}\n")
print(species_means)

# 2. Visual Proof: Scatter plot with Mystery Penguin highlighted as a Star
plt.figure(figsize=(9, 6))
sns.scatterplot(
    data=df,
    x="bill_length_mm",
    y="flipper_length_mm",
    hue="species",
    palette=["#2b5c8f", "#d95f02", "#7570b3"],
    s=70,
    alpha=0.6,
)

# Overlay mystery penguin
plt.scatter(
    mystery_penguin["bill_length_mm"],
    mystery_penguin["flipper_length_mm"],
    color="red",
    s=250,
    marker="*",
    edgecolor="black",
    linewidth=1.5,
    label="Mystery Penguin",
    zorder=5,
)

plt.title(
    "Mystery Penguin Identification Plot",
    fontsize=13,
    fontweight="bold",
    pad=15,
)
plt.xlabel("Bill Length (mm)", fontsize=11, fontweight="bold")
plt.ylabel("Flipper Length (mm)", fontsize=11, fontweight="bold")
plt.legend(frameon=True, facecolor="white")
plt.grid(True, linestyle="--", alpha=0.3)
sns.despine(top=True, right=True)

plt.tight_layout()
plt.show()
```

```
--- Mystery Penguin Values vs Species Means ---
Mystery Penguin: {'bill_length_mm': 46.0, 'bill_depth_mm': 15.0, 'flipper_length_mm': 217.0, 'body_mass_g': 5000.0}
```